

What is the right geometry for minimization of MMD?

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1 Introduction

Given a functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$, its gradient-flow dynamics depend on the choice of geometry on the underlying space of measures. One of the most commonly used geometries is the Wasserstein-2 geometry, induced by the Wasserstein-2 distance. Its associated Onsager operator, which is the inverse of the Riemannian metric tensor, is given by $\mathbb{K}_W(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto -\operatorname{div}(\rho \nabla \xi)$. Hence, the Wasserstein gradient flow of $\mathcal{F}$ is

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]). \quad (\text{W-GF})$$

Similarly, the Fisher–Rao geometry is induced by the Hellinger distance, with associated Onsager operator $\mathbb{K}_{\text{FR}}(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto \rho, \xi$. Consequently, the Fisher–Rao gradient flow of $\mathcal{F}$ is

$$\partial_t \mu_t = -\mu_t \left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{FR-GF})$$

An interpolation between these two dynamics is given by the well-known Wasserstein–Fisher–Rao gradient flow:

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mu_t \left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{WFR-GF})$$

Recently, a different geometry is considered in [Zhu \[2024\]](#), [Zhu and Mielke \[2024\]](#), [Gladin et al. \[2024\]](#), which uses the MMD as the geometry. The associated Onsager operator is $\mathbb{K}_{\text{MMD}}(\rho) = \mathbb{K}_{\text{FR}}(\rho) \circ \mathcal{K}_\rho^{-1}$. The Interaction-Force Transport (IFT) gradient flow is described as the following:

$$\partial_t \mu_t = -\mathbb{K}_{\text{MMD}}(\mu_t) \mathcal{F}'[\mu_t] = -\mu_t \mathcal{K}_{\mu_t}^{-1} \mathcal{F}'[\mu_t] = -\mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{IFT-GF})$$

Similarly as WFR-GF, an interpolation of WGF and IFT-GF can also be constructed

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{WIFT-GF})$$

2 $\mathcal{F} = \text{MMD}$ and geometry=MMD

We focus on MMD due to its interpretation as the worst-case integration error in the unit ball of a RKHS. If $\mathcal{F}$ is the MMD distance $\mathcal{F}(\cdot) = \text{MMD}^2(\cdot, \pi)$, then $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot) d(\mu - \pi) = \mathcal{K}(\mu - \pi)$. MSIP uses the WFG-GF, but here we focus on the WIFT-GF. The main reason is that, [Gladin et al. \[2024\]](#) has proved that $\text{MMD}^2(\mu_t, \pi) \leq e^{-2\beta t} \cdot \text{MMD}^2(\mu_0, \pi)$.

Following Gladin et al. [2024], a way to simulate the WIFT-GF. is via an weighted interacting particle system. Let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$ be a weighted empirical measure at iteration t , where the weights do not necessarily sum up to 1. Write $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$ and $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$. Denote $m_\pi = \int k(x, \cdot) d\pi$. First evolve the Wasserstein gradient term,

$$x_i^{(t+1)} = x_i^{(t)} - \tau\alpha \left[\sum_{j=1}^N w_j^{(t)} \nabla_2 k(x_j^{(t)}, x_i^{(t)}) - \nabla m_\pi(x_i^{(t)}) \right].$$

Then, evolve the IFT gradient term, by solving the following optimal weights: let $\eta = \beta\tau\alpha^{-1}$,

$$\mathbf{w}^{(t+1)} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \pi \right) + \frac{1}{2\eta} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \mu_N^{(t)} \right) \right\}.$$

Interestingly, letting $K^{(t+1)} = K(\mathbf{X}_{t+1}) \in \mathbb{R}^{N \times N}$, the weight problem is

$$\min_{\mathbf{w}} \left\{ \frac{1}{2} \mathbf{w}^\top K^{(t+1)} \mathbf{w} - m_\pi(\mathbf{X}_{t+1})^\top \mathbf{w} + \frac{1}{2\eta} (\mathbf{w} - \mathbf{w}^{(t)})^\top K^{(t+1)} (\mathbf{w} - \mathbf{w}^{(t)}) \right\},$$

By definition of kernel quadrature, there is $m_\pi(\mathbf{X}_{t+1}) = K^{(t+1)} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$. And the solution is

$$\mathbf{w}^{(t+1)} = \frac{1}{1+\eta} \mathbf{w}^{(t)} + \frac{\eta}{1+\eta} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$$

Theorem 2.1 (Monotone improvement over the initial kernel quadrature rule). *Let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$, $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$, and $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$. Define the MMD energy by*

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i}, \pi \right),$$

and let $\mathbf{w}_{\text{KQ}}(\mathbf{X}) = K(\mathbf{X})^{-1} m_\pi(\mathbf{X})$ denote the unconstrained kernel quadrature weights associated with the locations $\mathbf{X}$.

Assume that k is twice continuously differentiable, that $K(\mathbf{X}_t)$ is positive definite for every t , and that the weights remain uniformly positive, namely $w_i^{(t)} \geq \underline{w} > 0$ for every i, t . Assume moreover that, for every admissible $\mathbf{w}$, the map $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w})$ has an L_X -Lipschitz gradient on a set containing all iterates. If $\tau\alpha \leq 2\underline{w}/L_X$, then

$$\text{MMD}(\mu_N^{(t+1)}, \pi) \leq \text{MMD}(\mu_N^{(t)}, \pi) \quad (1)$$

for every $t \geq 0$.

Proof. For fixed $\mathbf{X}$ and $\mathbf{w}$, let $g_i(\mathbf{X}, \mathbf{w}) = \sum_{j=1}^N w_j \nabla_2 k(x_j, x_i) - \nabla m_\pi(x_i)$. Differentiating the MMD energy with respect to x_i and using symmetry of the kernel gives $\nabla_{x_i} \mathcal{E}(\mathbf{X}, \mathbf{w}) = w_i g_i(\mathbf{X}, \mathbf{w})$. Hence, the position update can be written as $x_i^{(t+1)} = x_i^{(t)} - \frac{\tau\alpha}{w_i^{(t)}} \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$. Thus, the

Wasserstein substep is a diagonally preconditioned gradient step for $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w}^{(t)})$.

By the L_X -smoothness of $\mathcal{E}(\cdot, \mathbf{w}^{(t)})$, the descent lemma gives

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \sum_{i=1}^N \left\langle \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}), x_i^{(t+1)} - x_i^{(t)} \right\rangle$$

$$\begin{aligned}
& + \frac{L_X}{2} \sum_{i=1}^N \left\| x_i^{(t+1)} - x_i^{(t)} \right\|^2 \\
& = \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) - \tau\alpha \sum_{i=1}^N \left(w_i^{(t)} - \frac{L_X \tau\alpha}{2} \right) \left\| g_i(\mathbf{X}_t, \mathbf{w}^{(t)}) \right\|^2.
\end{aligned} \tag{2}$$

Since $w_i^{(t)} \geq \underline{w}$ and $\tau\alpha \leq 2\underline{w}/L_X$, we have $w_i^{(t)} - L_X \tau\alpha/2 \geq 0$. Therefore, $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$.

We next consider the weight update. For fixed $\mathbf{X}$, the dependence of the MMD energy on $\mathbf{w}$ is quadratic: $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \mathbf{w}^\top K(\mathbf{X}) \mathbf{w} - m_\pi(\mathbf{X})^\top \mathbf{w} + C_\pi$, where C_π is independent of $\mathbf{w}$. Since $K(\mathbf{X})$ is positive definite, the unique minimizer is $\mathbf{w}_{\text{KQ}}(\mathbf{X}) = K(\mathbf{X})^{-1} m_\pi(\mathbf{X})$. Completing the square gives

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \mathcal{E}(\mathbf{X}, \mathbf{w}_{\text{KQ}}(\mathbf{X})) + \frac{1}{2} \|\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X})\|_{K(\mathbf{X})}^2, \tag{3}$$

where $\|v\|_{K(\mathbf{X})}^2 = v^\top K(\mathbf{X}) v$.

By the weight update, $\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) = (1 + \eta)^{-1} (\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))$. Consequently,

$$\begin{aligned}
& \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\
& = \frac{1}{(1 + \eta)^2} \left[\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \right].
\end{aligned} \tag{4}$$

Since $(1 + \eta)^{-2} \leq 1$ and $\mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$ minimizes $\mathbf{w} \mapsto \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w})$, it follows that $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)})$.

Combining the two substeps yields

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}). \tag{5}$$

Hence, $\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$ is nonincreasing. Since

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_i w_i \delta_{x_i}, \pi \right),$$

we obtain $\text{MMD}(\mu_N^{(t+1)}, \pi) \leq \text{MMD}(\mu_N^{(t)}, \pi)$.

Finally, if $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$, monotonicity implies

$$\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)}) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)).$$

Equivalently,

$$\text{MMD}^2 \left(\mu_N^{(t)}, \pi \right) \leq \text{MMD}^2 \left(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi \right). \tag{6}$$

Taking square roots proves (??). $\square$

References

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